

<ans> ****Exam Paper: Object Oriented Programming (OOP) (Total: 15 Marks)****

Section A: Short Answer Questions (Total: 15 Marks)**

Q1. Define the term "Encapsulation" in OOP. (2 Marks)**

****Answer:****

Encapsulation is an OOP concept that binds data (attributes) and methods (functions) into a single unit (class) while restricting direct access to some of the object's components. It helps in data hiding and protecting the internal state of an object.

Q2. What is Polymorphism? Explain with an example. (2 Marks)**

****Answer:****

Polymorphism allows objects of different classes to be treated as objects of a common superclass. It can be achieved through method overloading (compile-time) or method overriding (runtime).

****Example:****

```
```java
class Animal {
 void sound() { System.out.println("Animal sound"); }
}
class Dog extends Animal {
 @Override
 void sound() { System.out.println("Bark"); }
}
```
```

Q3. Differentiate between "Abstract Class" and "Interface" in Java. (3 Marks)**

****Answer:****

| **Abstract Class** | **Interface** |
|---------------------------|----------------------|
|---------------------------|----------------------|

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| | |
|---|--|
| Can have abstract and concrete methods. | Only abstract methods (before Java 8). |
|---|--|

| | |
|------------------------|------------------|
| Can have constructors. | No constructors. |
|------------------------|------------------|

| | |
|---|--------------------------------|
| Supports single and multiple inheritance. | Supports multiple inheritance. |
|---|--------------------------------|

| | |
|-------------------------------------|---|
| Variables are non-final by default. | Variables are always `public static final`. |
|-------------------------------------|---|

Q4. What is the "Diamond Problem" in Inheritance? How does

Java resolve it? (3 Marks)**

Answer:

The Diamond Problem occurs when a class inherits from two classes that have a common superclass, leading to ambiguity in method resolution. Java resolves it by:

- Not supporting multiple inheritance for classes (only interfaces).
- Using `default` methods in interfaces with explicit overriding.

Q5. Explain the concept of "Method Overloading" with an example. (3 Marks)

Answer:

Method overloading allows multiple methods with the same name but different parameters (type or number) in the same class.

Example:

```
```java
class Calculator {
 int add(int a, int b) { return a + b; }
 double add(double a, double b) { return a + b; }
}
```
```

Q6. What is the purpose of the `final` keyword in Java? (2 Marks)

Answer:

- When applied to a variable: Makes it a constant (immutable).
- When applied to a method: Prevents method overriding.
- When applied to a class: Prevents inheritance.

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